



Integrate footprint

Design a sidecut from its radius: draw the radius along the ski as sketch lines, get the outline -- exact arcs where the radius is constant, placed so the waist lands where you ask.

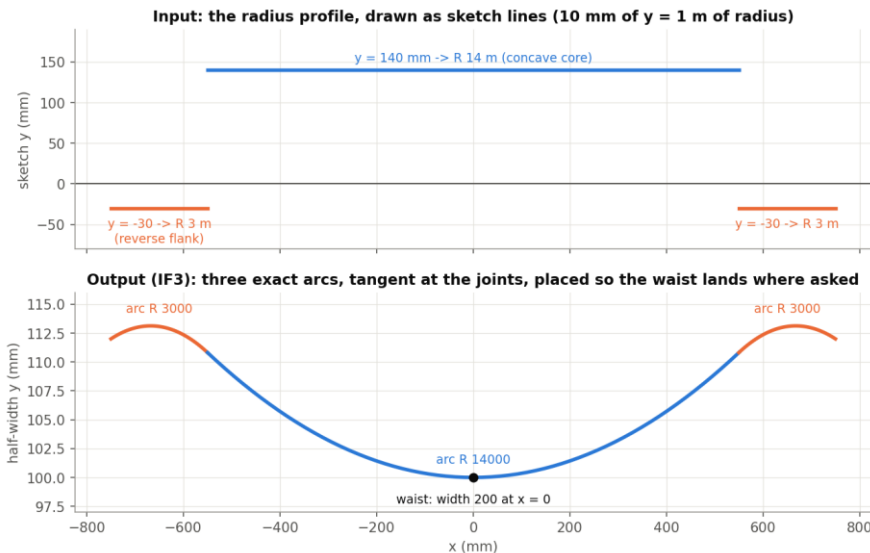


What it does

A sidecut is designed by its radii, but drawn as an outline. Integrate footprint turns a radius profile into the outline: curvature integrated twice, with the waist location (or taper) and width you set. Constant-radius stretches come out as true arcs that report their radius.

Use it when

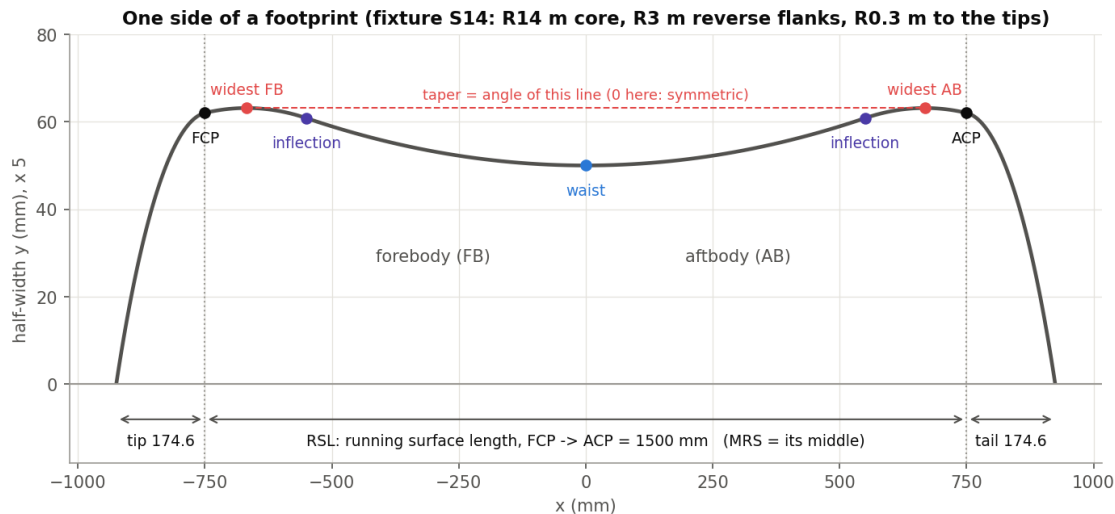
- Start a sidecut from radius targets
- Try a reverse-radius flank or a transition
- Need arcs that show 'R 14000' when clicked



IF3: profile lines at R14 m and R3 m (reverse) -> three exact, tangent arcs.



The words



Contacts and RSL

FCP / ACP where the ski meets the snow;
RSL between them, MRS its middle.

Waist, widest, inflection

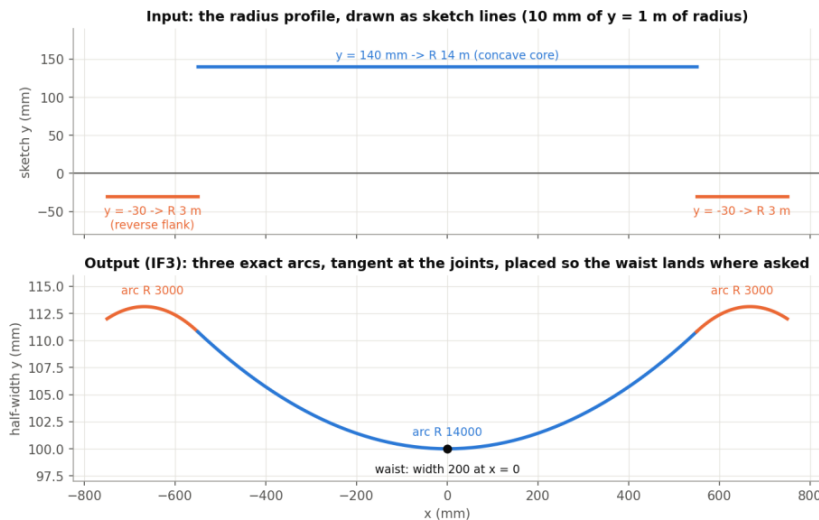
Narrowest point; widest point of each
half; where the curvature changes sign.

Taper

Angle of the line joining the widest
points; positive when the tip is wider.



Exact where it matters



Constant radius -> arc

Closed form $\sin(\phi) = \sin(\phi_0) + (x - x_0) / R$: an exact circle, emitted as a sketch arc.

Changing radius -> spline

A sloped profile line is integrated numerically and joins the arcs tangentially.

Placed by a solver

Tangent angle solved so the waist lands at the waist location (or the taper matches), then shifted to the waist width.



The dialog (1/2)

IMAGE MISSING:

docs/decks/integrate_footprint/shots/dialog.
png

#	Parameter	What it means
1	Spline method	APPROX / FIT: how transition splines are made.
2	Build Mode	One curve per same-sign region, or per profile edge.
3	Unify curves	One curve for all (ignored when there are arcs).
4	Strict	Force arcs / lines -- rational curves that do NOT report a radius.
5	Waist/Taper Angle Calculations	WAIST: place the narrowest point; TAPER: set the overall taper angle.
6	Waist location	World X of the waist (65 mm).
7	Waist width	Full width at the waist (95 mm).



The dialog (2/2)

IMAGE MISSING:

docs/decks/integrate_footprint/shots/dialog.
png

#	Parameter	What it means
8	Radius Profile(s)	The profile sketch lines: 10 mm of $y = 1$ m of radius; below $y = 0$ = reverse.
9	Contact points	FCP sets which end is the forebody (taper sign). ACP is not used yet.
10	Integration definition	Y-axis scaling: 10 / 20 / 50 / 100 mm per metre.
11	Spline approximation parameters	Degree, maximum control points, tolerance.
12	Debug & details	Samples per profile edge; group output into one wire.
13	Recalculate?	No effect.



Examples (Footprint tests)

IF1: one line at $y = 140$

One exact arc R14000, centre (0, 14100), waist width 200.

IF2: lines 25 / 14 / 25 m

Three exact arcs, tangent at the joints, waist (0, 100).

IF5: taper 0.25 deg

Placed by taper instead of waist location; end heights give exactly 0.25 deg.



Outputs and messages

Outputs

Key	What it holds
(wire)	One +Y side wire: sketch arcs for constant radius, splines for transitions.

Messages

	Message / what to do
(none)	No messages; a solver that does not converge stops on its last try silently.



Tips and limits



Horizontal lines for arcs

Only a constant-radius (horizontal) profile line gives an exact arc.



Profile edges

An edge may not cross $y = 0$: split concave and reverse regions.



Strict

Its arcs are rational splines and report no radius; leave it off for radius callouts.



Where it lives

Document	footprint (52724f3a857fa52d3ecceb77)
Tab	integrateFootprint (+ fpt_geometry, arcFit)
Tests	Footprint tests studio (build_ / check_footprint_tests.py): IF1-IF8
Explainer	docs/explainers/footprint/footprint_explained.md, section 1.2, 2.1
Related	Integrate footprint, Analyze footprint, Scale Footprint, Arc fit, Generate Footprint Points